Fares Mohamed

Alexandria, Egypt | faresmohamed260@gmail.com | github.com/faresmohamed260
linkedin.com/in/fares-mohamed-b83454194 | **Machine Learning Engineer – NLP, Generative AI, Computer Vision**

Summary

Computer Science graduate ranked first in class, building measured ML pipelines and multimodal systems from reproducible evaluation through durable web delivery, with a focus on NLP, retrieval, vision, and generative media.

Experience

Armstrong 2023–2025
AI & Automation Engineer / Technical Content Creator (Part-time)

- Combined n8n, custom GPTs, open-source models, scripts, and visual-generation tools to automate structured educational content for the Digital Egypt Cubs Initiative.
- Produced AI, robotics, Arduino, circuits, and programming curricula, demonstrations, guides, presentations, and videos.

Kayfa Aug–Dec 2025
AI Subject Matter Expert

- Authored ML lessons, assignments, assessments, and Python notebooks on neural networks, CNNs, clustering, anomaly detection, and unsupervised learning.

Engineering for Kids-Egypt 2023–2025
AI & Robotics Instructor (Seasonal)

- Taught students aged 11–16 AI, robotics, Python, C++, Arduino, and circuits through project-based programs.

Selected Projects

S.A.G.A. – Local-First Literary NLP
Python, NLP, FastAPI, Supabase/Postgres, BookNLP, LitBank

- Built evidence/provenance contracts and reproducible benchmarks for identity, dialogue, events, participants, relationships, temporal cues, and state candidates over long-form fiction.
- Achieved 0.8563 deterministic quote F1; benchmarked a 0.7791 event-trigger challenger and persistent BookNLP stdio with exact semantic equality and lower repeated-run latency.

RenderLab – Generative Media Platform
Next.js, Supabase, R2, ComfyUI, Modal, Playwright

- Delivered a production-domain closed beta for image/video generation and editing with durable asynchronous jobs, multi-reference inputs, model-aware parameters, failover, media lineage, and exact-head browser qualification.

Fares Uniform – Experimental Business Platform
Odoo, Python, Next.js, PostgreSQL

- Implemented and validated offline POS, preorders/payments, stock and B2B workflows plus a bilingual public site; 149 Odoo tests, 8/8 browser journeys, synthetic UAT, backup/restore, and observability passed.

VisionDeck – Computer Vision Suite
Python, OpenCV, MediaPipe, YOLO, Streamlit

- Developed real-time face, hand-gesture, and object-detection workflows for webcam and Android IP-camera feeds.

Education

Alexandria University, Faculty of Computer and Data Science 2022–2026
B.Sc. Computer Science (Intelligent Systems) | **Class Rank: 1st**

Skills

ML/AI: NLP, computer vision, RAG, hybrid retrieval, LLM orchestration, evaluation, diffusion models, scikit-learn
Engineering/Data: Python, C++, FastAPI, LangGraph, React, Docker, pytest, Playwright, PostgreSQL, Supabase, pgvector
Generative Media: ComfyUI, Qwen-Image, FLUX.2, ControlNet, LoRA, reference conditioning, Modal | **Languages:** Arabic, English

Certifications

Machine Learning Specialization | Deep Learning Specialization | Google Cloud Big Data & ML Fundamentals